Robert Dargavel Smith

Machine Learning Engineer & Quantitative Analyst
Authorized to work in the UK and EU

+34 614 50 99 57 teticio@gmail.com Madrid, Spain
in/attentioncoach github.com/teticio deej-ai.online

Summary

I am both a highly technical Individual Contributor and Manager, with many years of experience of building and leading teams and deploying models in production. I am passionate about sharing knowledge and collaboration, and by nature I am extremely persistent and driven by a need to understand how things work under the hood.

Tech Stack

Python - Hugging Face - PyTorch - TensorFlow - Docker - Git - Helm - Kubernetes - Terraform - Terragrunt - AWS (Lambda, EC2, SageMaker) - JavaScript - ReactJS - React Native - Selenium - Linux - `C++` - `C` - Elasticsearch

Professional Experience

Director Machine Learning Engineering, Clarity AI **London, UK** *4/2022 - present*

- Leading NLP / GenAI / MLE
  - Reporting to CTO, managing 10 people.
  - Lead successful hiring of 6 Data Scientists.
  - Leading all NLP projects (Information Extraction from PDFs, Controversies in news, Generative AI).
  - Created MLE team to bridge the gap with Engineering, focusing on tooling, implementation of best practices.
  - Gained trust of Engineering to manage cloud infrastructure thus increasing autonomy.
  - Ideated internal SaaS platform to empower SMEs to create models and decouple SME & DS workflows.
- Hands-on
  - Developed MCP Server for sale on AWS Marketplace, A2A agents, MS Teams and Discord bots.
  - SageMaker Studio for Teams. Developed IaC and tooling to replace existing ML platform, leading to $500K p.a. savings.
  - Deployed EC2 Slackbot to facilitate spinning up / monitoring / tearing down EC2 instances.
  - Ethical web-scraping. Employed techniques to obtain data for POCs, including custom LLM web agents.
  - Llama (LoRA). Fine-tuned LLMs on specialized tasks and researched reasoning tokens.
  - Policy detection. Improved HITL process and fine-tuning of BERT models, trained to detect compliance with policies.
  - OpenAI Proxy. Deployed API gateway for fine-grained monitoring / control of OpenAI usage by project.

Head of Data Science, IHS Markit (now roughly one third of S&P Global) **London, UK** *4/2020 - 4/2022*

- Reported directly into Executive Vice President group CTO of IHS Markit (16,000 employees).
- Managed Centre of Excellence of 20 Data Scientists based in London, Gdansk and Bangalore.
- Collaborated with all business verticals (Financial Services, Automotive, Energy, Maritime & Trade).

Managing Director, Santander Group **Madrid, Spain** *1/2001 - 3/2020*

- Several roles including Global Head of Quantitative Analysis and Head of CVA Desk.
- Built and led team of over 20 people from more than 10 nationalities based in Madrid, London & New York.
- Developed and deployed models in `C++` and Python used to price, hedge and value the bank's derivative portfolios.
- Started and ran CVA Desk, responsible for managing several hundred million euros of P&L risk.
- Frequent speaker at industry conferences in Europe, USA & Asia.

Quantitative Analyst, ABN Amro **London, UK** *1995 - 1/2001*

- Developed models in `C` for pricing, hedging and valuing equity derivatives and convertible bonds.

Software Engineer, Harlequin (now Global Graphics) **Cambridge, UK** *1993 - 1995*

- Developed device drivers for commercial printing presses.

Selected Projects

- Audio diffusion. Music generation, included in Hugging Face's Diffusers package.
- Deej-AI. Playlist generation with a Deep Convolutional Neural Network trained on audio samples.
- deej-ai.online-app. ReactJS / React Native app deployed with Kubernetes.
- Lambda scraper. AWS Lambda functions as a proxy pool for web scraping.
- Python Envy. VS Code extension to automatically activate Python environments when navigating a monorepo.
- LALAL.AI. Collaboration with a startup specializing in stemming audios with ML.

Academic Experience

Professor, MBIT School **Madrid, Spain** *2020*

Advanced Machine Learning Expert Programme, specializing in Deep Neural Networks and NLP.

Professor, Universidad de Alcalá **Madrid, Spain** *2013 - 2020*

Masters of Artificial Intelligence and Masters of Quantitative Finance.

Publications

Landmarks in XVA: From Counterparty Risk to Funding Costs and Capital, Risk Books *2016*

Chapter on "Managing XVA in the Ring Fenced Bank".

An Almost Exact Simulation Method for the Heston Model, Journal of Computational Finance *2007*

Education

Advanced Machine Learning Expert Programme, MBIT School **Madrid, Spain** *2019*

Awarded maximum grade and was invited to be a Professor in subsequent editions of the course.

BA First Class Honors in Mathematics Pembroke College, University of Oxford **Oxford, UK** *1990-1993*

Awarded Pembroke College prize for Mathematics.

Fun facts

- Actor in several theatre productions (Talkin' Turkey Theare Company, Carlton Theatre Group, Honda Teatro).
- Certified Mindfulness (MBCT) Instructor.
- Marathon runner (PB 2:47 New York City, 2013).
- Ironman Finisher (Florianopolis, Brazil, 2011).
- DJ (club, radio, produced 12" single featured on Gilles Peterson compilation).
- Bronze medalist in World Rowing Junior Championships (coxed four, 1990).

Natural Languages

- **English** [Native]
- **Spanish** [Bilingual]